

Vanishing of the unipotent generators at zero in the presented group $G(\mathfrak{m})$

Abstract

We prove Lemma 3.1 (book label "gsimple zero generators trivial"). It states that in the group $G(\mathfrak{m})$ given by generators and relations, each unipotent generator $X_{-1}(0)$, $Y_{-1}(0)$, $X_{\ell,jk}(0)$, $Y_{\ell,jk}(0)$ is the identity. In the formalization, the coefficient function c of the modular function J is replaced by an arbitrary function $c: \mathbb{N} \rightarrow \mathbb{N}$, and the result holds for every such c . The proof has two steps. First, a general lemma: a map from $(\mathbb{C}, +)$ to a group that turns sums into products sends 0 to 1. Second, four lemmas showing that each family of unipotent generators has this property, by the additivity relations (Re:1a), (Re:1b), (Im:1a) and (Im:1b). The main theorem and all lemmas are machine-verified.

Source	Carbone et al., G-Simple (draft), GSimple_Lemma_3_1
Status	proof machine-verified (Lean 4, Mathlib)
Lemmas	5
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1 Setting

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$, $\mathbb{C}^\times$ is the group of units of $\mathbb{C}$, and $c: \mathbb{N} \rightarrow \mathbb{N}$ is an arbitrary function. The function c plays the role of the coefficient function of J ; no property of c is assumed.

Definition 1.1 (Index set). Let

$$E_c = \{(\ell, j, k) \in \mathbb{N}^3 \mid 1 \leq k \leq c(j), \ell < j\}.$$

Since $\ell < j$, every $(\ell, j, k) \in E_c$ has $j \geq 1$.

For $e = (\ell, j, k) \in E_c$, put $e^* = (j - 1 - \ell, j, k)$. Then $e^* \in E_c$, because $j - 1 - \ell < j$.

Definition 1.2 (Structure constants). For $\ell, j \in \mathbb{N}$, let

$$c_{\ell j} = (-1)^{\ell+1} \binom{j-1}{\ell} (\ell+1)(j-\ell) \in \mathbb{C}.$$

For $\ell < j$ all differences here are ordinary differences of natural numbers, and $c_{\ell j} \neq 0$. Divisions by $c_{\ell j}$ below are therefore genuine divisions in $\mathbb{C}$.

Definition 1.3 (Generators). Let $\mathcal{X}_c$ be the set of formal symbols

$$H_1(s), H_2(s) \ (s \in \mathbb{C}^\times), \quad X_{-1}(u), Y_{-1}(u) \ (u \in \mathbb{C}), \quad X_e(u), Y_e(u) \ (e \in E_c, u \in \mathbb{C}).$$

For $e = (\ell, j, k)$ we also write $X_{\ell, jk}(u) = X_e(u)$ and $Y_{\ell, jk}(u) = Y_e(u)$. Let $F(\mathcal{X}_c)$ be the free group on $\mathcal{X}_c$.

In $F(\mathcal{X}_c)$ define, for $s \in \mathbb{C}^\times$ and $e = (\ell, j, k) \in E_c$:

$$\begin{aligned} \tilde{w}_{-1}(s) &= X_{-1}(s) Y_{-1}(-s^{-1}) X_{-1}(s), & \tilde{w}_{-1} &= \tilde{w}_{-1}(1), \\ \tilde{w}_e(s) &= X_e(s) Y_e\left(\frac{-s^{-1}}{c_{\ell j}}\right) X_e(s), & \tilde{w}_e &= \tilde{w}_e(1). \end{aligned}$$

The commutator is $(a, b) = aba^{-1}b^{-1}$.

Definition 1.4 (Relators). Let $\mathcal{R}_c \subseteq F(\mathcal{X}_c)$ consist of the following elements. In each list we write a relation “ $a = b$ ”, which stands for the relator ab^{-1} . A relation “ $(a, b) = 1$ ” stands for the relator (a, b) . The variables range over $s, t \in \mathbb{C}^\times$, $u, v \in \mathbb{C}$, and $e = (\ell, j, k)$, $f = (m, p, q) \in E_c$.

- (H:1a) $H_1(s)H_1(t) = H_1(st)$.
- (H:1b) $H_2(s)H_2(t) = H_2(st)$.
- (H:2) $H_1(s)H_2(t) = H_2(t)H_1(s)$.
- (Re:1a) $X_{-1}(u)X_{-1}(v) = X_{-1}(u+v)$.
- (Re:1b) $Y_{-1}(u)Y_{-1}(v) = Y_{-1}(u+v)$.
- (Re:2) $Y_{-1}(-t)X_{-1}(s)Y_{-1}(t) = X_{-1}(-t^{-1})Y_{-1}(-t^2s)X_{-1}(t^{-1})$.
- (Re:3) $\tilde{w}_{-1}(s)\tilde{w}_{-1} = H_1(-s)H_2(-s^{-1})$.
- (Re:4a) $\tilde{w}_{-1}X_{-1}(u)\tilde{w}_{-1}^{-1} = Y_{-1}(-u)$.
- (Re:4b) $\tilde{w}_{-1}Y_{-1}(u)\tilde{w}_{-1}^{-1} = X_{-1}(-u)$.
- (Re:5a) $\tilde{w}_{-1}H_1(s)\tilde{w}_{-1}^{-1} = H_2(s)$.
- (Re:5b) $\tilde{w}_{-1}H_2(s)\tilde{w}_{-1}^{-1} = H_1(s)$.
- (Re:6a) $H_1(s)X_{-1}(u)H_1(s)^{-1} = X_{-1}(su)$.
- (Re:6b) $H_2(s)X_{-1}(u)H_2(s)^{-1} = X_{-1}(s^{-1}u)$.
- (Re:6c) $H_1(s)Y_{-1}(u)H_1(s)^{-1} = Y_{-1}(s^{-1}u)$.
- (Re:6d) $H_2(s)Y_{-1}(u)H_2(s)^{-1} = Y_{-1}(su)$.

- (Im:1a) $X_e(u)X_e(v) = X_e(u+v)$.
- (Im:1b) $Y_e(u)Y_e(v) = Y_e(u+v)$.
- (Im:2) $Y_e(-t)X_e(s)Y_e(t) = X_e\left(\frac{-t^{-1}}{c_{\ell j}}\right)Y_e(-c_{\ell j}t^2s)X_e\left(\frac{t^{-1}}{c_{\ell j}}\right)$.
- (Im:3) $\tilde{w}_e(s^{(\ell+1)(j-\ell)})\tilde{w}_e = H_1((-s)^{j-\ell})H_2((-s)^{\ell+1})$.
- (Im:4a) $\tilde{w}_eX_e(u)\tilde{w}_e^{-1} = Y_e(-u/c_{\ell j})$.
- (Im:4b) $\tilde{w}_eY_e(u)\tilde{w}_e^{-1} = X_e(-c_{\ell j}u)$.
- (Im:5a) $\tilde{w}_eH_1(s^{j-\ell})\tilde{w}_e^{-1} = H_2(s^{-(\ell+1)})$.
- (Im:5b) $\tilde{w}_eH_2(s^{\ell+1})\tilde{w}_e^{-1} = H_1(s^{-(j-\ell)})$.
- (Im:6a) $H_1(s)X_e(u)H_1(s)^{-1} = X_e(s^{\ell+1}u)$.
- (Im:6b) $H_2(s)X_e(u)H_2(s)^{-1} = X_e(s^{j-\ell}u)$.
- (Im:6c) $H_1(s)Y_e(u)H_1(s)^{-1} = Y_e(s^{-(\ell+1)}u)$.
- (Im:6d) $H_2(s)Y_e(u)H_2(s)^{-1} = Y_e(s^{-(j-\ell)}u)$.
- (U:1a) $(X_{-1}(u), X_e(v)) = 1$ if $\ell+1 = j$.
- (U:1b) $(Y_{-1}(u), Y_e(v)) = 1$ if $\ell+1 = j$.
- (U:1c) $(Y_{-1}(u), X_e(v)) = 1$ if $\ell = 0$.
- (U:1d) $(X_{-1}(u), Y_e(v)) = 1$ if $\ell = 0$.
- (U:2) $(X_e(u), Y_f(v)) = 1$ if $j \neq p$, or $k \neq q$, or $|\ell - m| > 1$.
- (U:3a) $(X_e(u), Y_f(v)) = X_{-1}(uv)$ if $e = (1, 2, k)$ and $f = (0, 2, k)$ for the same k .
- (U:3b) $(X_e(u), Y_f(v)) = Y_{-1}(uv)$ if $e = (0, 2, k)$ and $f = (1, 2, k)$ for the same k .
- (U:4a) $\tilde{w}_{-1}X_e(u)\tilde{w}_{-1}^{-1} = X_{e^*}((-1)^{j-\ell-1}u)$.
- (U:4b) $\tilde{w}_{-1}Y_e(u)\tilde{w}_{-1}^{-1} = Y_{e^*}((-1)^{j-\ell-1}u)$.

Definition 1.5 (The group). Let $G_c(\mathbf{m}) = F(\mathcal{X}_c)/N_c$, where N_c is the normal closure of $\mathcal{R}_c$. Let $\pi: F(\mathcal{X}_c) \rightarrow G_c(\mathbf{m})$ be the quotient homomorphism.

By construction, $\pi(r) = 1$ for every $r \in \mathcal{R}_c$. Consequently, if $ab^{-1} \in \mathcal{R}_c$, then $\pi(a) = \pi(b)$.

Theorem 1.6. *Let $c: \mathbb{N} \rightarrow \mathbb{N}$ be any function. Then in $G_c(\mathbf{m})$:*

- (a) $\pi(X_{-1}(0)) = 1$;
- (b) $\pi(Y_{-1}(0)) = 1$;
- (c) $\pi(X_e(0)) = 1$ for every $e \in E_c$;
- (d) $\pi(Y_e(0)) = 1$ for every $e \in E_c$.

This is Lemma 3.1 of the source for an arbitrary coefficient function c . Taking $c(j)$ to be the coefficients of J recovers the source statement. The hypothesis $c(2) \geq 1$, mentioned in the source's formalization note, is not needed here.

2 Lemmas

2.1 A general fact about additive families

Lemma 2.1 (Additive families vanish at zero). *Let G be a group and $f: \mathbb{C} \rightarrow G$ a map with $f(u+v) = f(u)f(v)$ for all $u, v \in \mathbb{C}$. Then $f(0) = 1$.*

Proof. Taking $u = v = 0$ gives $f(0) = f(0+0) = f(0)f(0)$. Hence

$$f(0) = f(0)(f(0)f(0)^{-1}) = (f(0)f(0))f(0)^{-1} = f(0)f(0)^{-1} = 1. \quad \square$$

2.2 Additivity of the unipotent families in $G_c(\mathfrak{m})$

Lemma 2.2 (Additivity of X_{-1}). *For every $c: \mathbb{N} \rightarrow \mathbb{N}$ and all $u, v \in \mathbb{C}$, we have $\pi(X_{-1}(u+v)) = \pi(X_{-1}(u))\pi(X_{-1}(v))$ in $G_c(\mathfrak{m})$.*

Proof. By (Re:1a), the element $X_{-1}(u)X_{-1}(v)X_{-1}(u+v)^{-1}$ lies in $\mathcal{R}_c$, so its image under π is 1. Since π is a homomorphism, this reads

$$\pi(X_{-1}(u))\pi(X_{-1}(v))\pi(X_{-1}(u+v))^{-1} = 1.$$

Multiplying on the right by $\pi(X_{-1}(u+v))$ gives the claim. $\square$

Lemma 2.3 (Additivity of Y_{-1}). *For every $c: \mathbb{N} \rightarrow \mathbb{N}$ and all $u, v \in \mathbb{C}$, we have $\pi(Y_{-1}(u+v)) = \pi(Y_{-1}(u))\pi(Y_{-1}(v))$ in $G_c(\mathfrak{m})$.*

Proof. By (Re:1b), the element $Y_{-1}(u)Y_{-1}(v)Y_{-1}(u+v)^{-1}$ lies in $\mathcal{R}_c$, so its image under π is 1. Hence

$$\pi(Y_{-1}(u)Y_{-1}(v))\pi(Y_{-1}(u+v))^{-1} = 1.$$

Multiplying on the right by $\pi(Y_{-1}(u+v))$ gives $\pi(Y_{-1}(u)Y_{-1}(v)) = \pi(Y_{-1}(u+v))$. Since π is a homomorphism, the left side equals $\pi(Y_{-1}(u))\pi(Y_{-1}(v))$. $\square$

Lemma 2.4 (Additivity of X_e). *For every $c: \mathbb{N} \rightarrow \mathbb{N}$, every $e \in E_c$ and all $u, v \in \mathbb{C}$, we have $\pi(X_e(u+v)) = \pi(X_e(u))\pi(X_e(v))$ in $G_c(\mathfrak{m})$.*

Proof. By (Im:1a), the element $(X_e(u)X_e(v))X_e(u+v)^{-1}$ lies in $\mathcal{R}_c$, so $\pi(X_e(u)X_e(v)) = \pi(X_e(u+v))$. Since π is a homomorphism,

$$\pi(X_e(u+v)) = \pi(X_e(u)X_e(v)) = \pi(X_e(u))\pi(X_e(v)). \quad \square$$

Lemma 2.5 (Additivity of Y_e). *For every $c: \mathbb{N} \rightarrow \mathbb{N}$, every $e \in E_c$ and all $u, v \in \mathbb{C}$, we have $\pi(Y_e(u+v)) = \pi(Y_e(u))\pi(Y_e(v))$ in $G_c(\mathfrak{m})$.*

Proof. By (Im:1b), the element $(Y_e(u)Y_e(v))Y_e(u+v)^{-1}$ lies in $\mathcal{R}_c$, so $\pi(Y_e(u)Y_e(v)) = \pi(Y_e(u+v))$. Since π is a homomorphism,

$$\pi(Y_e(u+v)) = \pi(Y_e(u)Y_e(v)) = \pi(Y_e(u))\pi(Y_e(v)). \quad \square$$

3 Proof of the theorem

Proof of Theorem 1.6. Fix c . Each part applies Lemma 2.1 with $G = G_c(\mathfrak{m})$ to a suitable map $f: \mathbb{C} \rightarrow G_c(\mathfrak{m})$.

- (a) Take $f(u) = \pi(X_{-1}(u))$. This map is additive by Lemma 2.2, hence $\pi(X_{-1}(0)) = 1$.
- (b) Take $f(u) = \pi(Y_{-1}(u))$. This map is additive by Lemma 2.3, hence $\pi(Y_{-1}(0)) = 1$.
- (c) Fix $e \in E_c$ and take $f(u) = \pi(X_e(u))$. This map is additive by Lemma 2.4, hence $\pi(X_e(0)) = 1$.
- (d) Fix $e \in E_c$ and take $f(u) = \pi(Y_e(u))$. This map is additive by Lemma 2.5, hence $\pi(Y_e(0)) = 1$. $\square$

Relation to the source proof. The source proof treats only part (a). It sets $u = v = 0$ in (Re:1a) to get $X_{-1}(0)^{-1}X_{-1}(0)X_{-1}(0) = 1$, and states that (b), (c), (d) follow in the same way from (Re:1b), (Im:1a) and (Im:1b).

The argument here is the same in substance, organised in two steps. First, it extracts from each additivity relation the statement that $u \mapsto \pi(\cdot(u))$ is a homomorphism $(\mathbb{C}, +) \rightarrow G_c(\mathfrak{m})$. Then it applies, in all four cases, the general fact that such a map sends 0 to 1. All four parts are carried out explicitly.

The only other difference is the one noted after the theorem. The coefficient function c is an arbitrary parameter, and no hypothesis on c is used, since only the relations (Re:1a), (Re:1b), (Im:1a) and (Im:1b) enter the proof.

A Correspondence with the formal proof

Each lemma of the text and the Lean declaration that states it; Theorem 1.6 is `gsimple_zero_generators_trivial`.

Lemma	Lean declaration	Status
Lemma 2.1	<code>add_zero_group_lemma</code>	verified
Lemma 2.2	<code>mk_xm_add</code>	verified
Lemma 2.3	<code>mk_ym_add</code>	verified
Lemma 2.4	<code>mk_xx_add</code>	verified
Lemma 2.5	<code>mk_yy_add</code>	verified